

AMC 10 Crash Course • Block 1 • The Toolkit

An optional deep dive after the block: six techniques, twelve problems at #12–18 with full solutions. For after the short course and T1.

This section is NOT part of the block's core 7.5 hours. It is for the moment when the short course is done, Test T1 is passed, and you want to push algebra deeper. Ninety minutes of worked problems: six techniques, 15 minutes each, two problems at AMC #12–18 level per technique, fully solved. How to use it: an honest attempt first (5–7 minutes per problem), then the solution — reading the solution before trying defeats the purpose.

T1. Symmetry and Substitution (15 min)

If an expression is unchanged by the swap $x \rightarrow 1/x$ or by exchanging $x \leftrightarrow y$, do not attack it head-on: introduce symmetric variables instead — $s = x + 1/x$ for power chains, $s = x + y$ and $p = xy$ for symmetric systems. Signals in the statement: pairs $x^n + 1/x^n$, expressions where x and y enter identically, a question that asks for a combination of the unknowns rather than the unknowns themselves.

PROBLEM 1 • POWER CHAIN

A real number x satisfies $x + 1/x = 3$. What is $x^5 + 1/x^5$?

→ The dead end that beckons: solve the quadratic $x^2 - 3x + 1 = 0$, get $x = (3 \pm \sqrt{5})/2$, and raise the irrationality to the fifth power. That is ten minutes and an almost guaranteed arithmetic slip. The trick: write $s_n = x^n + 1/x^n$. Multiplying by s_1 gives a recursion: $s_n \cdot s_1 = s_{n+1} + s_{n-1}$, that is $s_{n+1} = 3s_n - s_{n-1}$. Start: $s_1 = 3$; $s_2 = 3^2 - 2 = 7$. Then walk the chain: $s_3 = 3 \cdot 7 - 3 = 18$; $s_4 = 3 \cdot 18 - 7 = 47$; $s_5 = 3 \cdot 47 - 18 = \mathbf{123}$. Four mental multiplications and not a radical in sight.

PROBLEM 2 • SYMMETRIC SYSTEM

Real numbers x and y satisfy $x + y + xy = 11$ and $x^2y + xy^2 = 30$. What is the sum of all possible values of $x^2 + y^2$?

→ The dead end: solve the first equation for y and substitute into the second — a quartic appears. The trick: both equations are symmetric, so introduce $s = x + y$, $p = xy$. First equation: $s + p = 11$. Second: $x^2y + xy^2 = xy(x + y) = sp = 30$. So s and p are the roots of $t^2 - 11t + 30 = 0$, that is $\{5, 6\}$: either $s = 5$, $p = 6$, or $s = 6$, $p = 5$. Check that real x , y exist: we need $s^2 \geq 4p$; for $(5, 6)$: $25 \geq 24$ — yes; for $(6, 5)$: $36 \geq 20$ — yes. Then $x^2 + y^2 = s^2 - 2p$: $25 - 12 = 13$ in the first case and $36 - 10 = 26$ in the second. The sum of the possible values is $13 + 26 = \mathbf{39}$.

T2. Telescoping and Grouping (15 min)

A long sum or product with an ellipsis is almost never about patience — it is about collapse: each term splits into a difference (or ratio) of neighboring quantities, and the middle cancels itself out. Signals: denominators of the form $n(n + k)$, factors like $1 - 1/n^2$, a sum with “...” spanning dozens of terms. The first move is always the same: write out two or three terms and watch what cancels.

PROBLEM 1 • TELESOPING SUM

What is the sum $\frac{1}{1 \cdot 3} + \frac{1}{2 \cdot 4} + \frac{1}{3 \cdot 5} + \dots + \frac{1}{24 \cdot 26}$?

→ The dead end: put 24 fractions over a common denominator. The trick: split the general term. From $\frac{1}{n(n+2)} = \frac{1}{2}(\frac{1}{n} - \frac{1}{n+2})$ the sum becomes $\frac{1}{2}(1 - \frac{1}{3} + \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{5} + \dots)$. The gap is 2, so TWO leading positive terms and TWO trailing negative terms survive — the classic error is assuming “everything but the first and last” cancels. What remains is $\frac{1}{2}(1 + \frac{1}{2} - \frac{1}{25} - \frac{1}{26}) = \frac{1}{2} \cdot \frac{462}{325} = \frac{231}{325}$. Sanity check: the answer is slightly under $3/4$ — exactly right, since the full infinite sum equals $3/4$.

PROBLEM 2 • COLLAPSING PRODUCT

What is the product $(1 - \frac{1}{2^2})(1 - \frac{1}{3^2})(1 - \frac{1}{4^2}) \cdot \dots \cdot (1 - \frac{1}{50^2})$?

→ The dead end: multiply decimal approximations — 49 factors will not yield to brute force. The trick: each factor is a difference of squares: $1 - \frac{1}{k^2} = \frac{(k-1)(k+1)}{k^2}$. The product splits into two running chains: $\frac{1 \cdot 3}{2 \cdot 2} \cdot \frac{2 \cdot 4}{3 \cdot 3} \cdot \frac{3 \cdot 5}{4 \cdot 4} \cdot \dots \cdot \frac{49 \cdot 51}{50 \cdot 50}$. The “ k ” chain gives $1 \cdot 2 \cdot 3 \cdot \dots \cdot 49$, the “ $k+1$ ” chain gives $3 \cdot 4 \cdot \dots \cdot 51$, and the denominator is $(2 \cdot 3 \cdot \dots \cdot 50)^2$. Nearly everything cancels; only the edges survive: $\frac{1}{2} \cdot \frac{51}{50} = \frac{51}{100}$. Remember the general shape: up to n the product is $(n+1)/(2n)$ — just over one half, which is an instant sanity check.

T3. Difference of Squares and Clever Factoring (15 min)

An integer equation with a product of variables, or a huge number of the form $a^4 + 4b^4$, signals that factoring — not computing — solves the problem. Two workhorses: Simon’s Favorite Factoring Trick (add a constant so that $xy + ax + by$ folds into a product of two binomials) and the Sophie Germain identity: $a^4 + 4b^4 = (a^2 + 2b^2 + 2ab)(a^2 + 2b^2 - 2ab)$. Recognize the shape, then complete it.

PROBLEM 1 • SIMON’S TRICK

How many ordered pairs of positive integers (x, y) satisfy $xy + 3x + 2y = 42$?

→ The dead end: solve for $y = (42 - 3x)/(x + 2)$ and test x one by one, checking divisibility each time. It works, but slowly and with no guarantee of completeness. Simon’s trick: group $x(y + 3) + 2y$ and add 6 to close the second binomial: $xy + 3x + 2y + 6 = (x + 2)(y + 3) = 48$. Now it is a divisor problem. Since $x \geq 1$ and $y \geq 1$, we need divisors of 48 with $x + 2 \geq 3$ and $y + 3 \geq 4$, that is $3 \leq x + 2 \leq 12$. Divisors of 48: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48; those that qualify: 3, 4, 6, 8, 12. They give (1, 13), (2, 9), (4, 5), (6, 3), (10, 1) — **5** pairs in all. Check one: $4 \cdot 5 + 12 + 10 = 42$. Correct.

PROBLEM 2 • SOPHIE GERMAIN

Find the largest prime factor of $20^4 + 4 \cdot 9^4$.

→ The dead end: compute $160,000 + 26,244 = 186,244$ and trial-divide by primes — you would need on the order of a hundred trials to reach the right one. The trick: the shape $a^4 + 4b^4$ is the Sophie Germain identity with $a = 20$, $b = 9$ (the identity itself is a disguised difference of squares: $a^4 + 4b^4 = (a^2 + 2b^2)^2 - (2ab)^2$). Evaluate the pieces: $a^2 + 2b^2 = 400 + 162 = 562$, $2ab = 360$. So the number equals $(562 + 360)(562 - 360) = 922 \cdot 202$. Factor further: $922 = 2 \cdot 461$, $202 = 2 \cdot 101$. The number 461 is divisible by none of 2, 3, 5, 7, 11, 13, 17, 19 (and $21^2 > 461$), so it is prime; 101 is prime too. Altogether $186,244 = 2^2 \cdot 101 \cdot 461$, and the largest prime factor is **461**.

T4. Vieta Beyond the Basics (15 min)

Vieta’s formulas are more than “sum and product of the roots.” From $r + s$ and rs you can assemble any symmetric combination (squares, cubes, reciprocals), build a new quadratic whose roots are shifted or reciprocal, and turn the demand “the roots are integers” into a divisor problem. Signals: the question asks for an expression in the roots rather than the roots; the equation has a parameter and asks for which values the roots are integers.

PROBLEM 1 • A NEW QUADRATIC FROM AN OLD ONE

Let r and s be the roots of $x^2 - 6x + 2 = 0$. The quadratic $x^2 + bx + c = 0$ has roots r^2 and s^2 . What is $b + c$?

→ The dead end: find the roots explicitly, $r, s = 3 \pm \sqrt{7}$, square them and assemble the equation out of irrationalities. The trick: the new quadratic is built from the sum and product of the new roots, and those come from the old ones via Vieta. Old: $r + s = 6$, $rs = 2$. New sum: $r^2 + s^2 = (r + s)^2 - 2rs = 36 - 4 = 32$. New product: $r^2 s^2 = (rs)^2 = 4$. By Vieta for the new equation, $b = -32$ (MINUS the sum!) and $c = 4$. The sign is the trap: half of all wrong answers here are $+32$. So $b + c = -32 + 4 = \mathbf{-28}$.

PROBLEM 2 • INTEGER ROOTS AND DIVISORS

For how many integer values of k does the equation $x^2 - kx + 36 = 0$ have two integer roots (not necessarily distinct)?

→ The dead end: demand that the discriminant $k^2 - 144$ be a perfect square and scan values of k — a correct condition, but the scan is long and the negative values are easy to lose. The trick: by Vieta, integer roots r, s give $rs = 36$ and $k = r + s$, so k is the sum of a same-sign divisor pair of 36. Positive pairs: (1, 36), (2, 18), (3, 12), (4, 9), (6, 6) — sums 37, 20, 15, 13, 12: five values. Both roots may also be negative (the product is still 36!), giving five more: -37, -20, -15, -13, -12. All ten sums are distinct, so there are exactly **10** values of k . Discriminant check: $k = 13$ gives $169 - 144 = 25 = 5^2$. It fits.

T5. Estimation and Comparison Without Computing (15 min)

When the numbers are plainly uncomputable — powers with exponent 40, sums of a hundred radicals — the problem is asking not for a value but for a comparison or two neighboring bounds. Tools: bring powers to a common exponent (compare bases) or a common base (compare exponents); squeeze a sum between two “staircases” built from differences of radicals. Signals: “which is larger,” “between which integers,” “what is the integer part.”

PROBLEM 1 • COMMON EXPONENT

Arrange the numbers $A = 2^{60}$, $B = 3^{40}$, $C = 10^{18}$ in increasing order.

→ The dead end: estimate digit counts via logarithms you do not have at the AMC, or worse, try to multiply things out. The trick: 60 and 40 share the divisor 20 — bring both to a common exponent: $A = (2^3)^{20} = 8^{20}$, $B = (3^2)^{20} = 9^{20}$. Base 9 beats base 8, so $B > A$. Now C : the exponent 18 does not match 20, but $2^{60} = (2^{10})^6 = 1024^6$ while $10^{18} = (10^3)^6 = 1000^6$. Since $1024 > 1000$, we get $A > C$. Conclusion: **$C < A < B$** , that is $10^{18} < 2^{60} < 3^{40}$. The whole solution is two rewritings of powers — not a single large number computed.

PROBLEM 2 • SQUEEZING A SUM OF RADICALS

What is the integer part of the sum $S = \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{100}}$?

→ The dead end: add a hundred irrational terms to two decimal places — impossible in the three minutes you have. The trick: squeeze each term between differences of radicals. Multiplying by the conjugate gives $2(\sqrt{k+1} - \sqrt{k}) = \frac{2}{\sqrt{k+1} + \sqrt{k}} < \frac{1}{\sqrt{k}} < \frac{2}{\sqrt{k} + \sqrt{k-1}} = 2(\sqrt{k} - \sqrt{k-1})$. Both sides telescope! Lower bound: $S > 2(\sqrt{101} - \sqrt{1}) > 2(10.05 - 1) = 18.1$. Upper bound (keep the first term, 1, as is and bound the rest): $S < 1 + 2(\sqrt{100} - \sqrt{1}) = 1 + 18 = 19$. So $18.1 < S < 19$, and the integer part is **18**. The true value is $S \approx 18.59$ — safely inside the trap we built.

T6. Equations with Built-in Constraints (15 min)

Sometimes the main tool is not a transformation but the constraint itself: the number of solutions of an absolute-value equation jumps as a parameter moves; a linear relation turns a two-variable expression into a one-variable quadratic. Signals: “for which parameter values are there exactly N solutions,” “the greatest value of xy given that...” The plan is always the same: substitute the constraint, reduce to one variable, examine the borderline cases.

PROBLEM 1 • ABSOLUTE VALUE WITH A PARAMETER

For what value of the parameter a does the equation $||x - 4| - 3| = a$ have exactly three solutions?

→ The dead end: expand four sign combinations and solve each for arbitrary a — a porridge of cases with no view of the whole. The trick: count solutions layer by layer, watching for the branch that degenerates. For $a < 0$ there are no solutions. For $a \geq 0$ the outer layer gives $|x - 4| = 3 + a$ or $|x - 4| = 3 - a$. First branch: $3 + a > 0$ always — exactly 2 solutions. Second branch: 2 solutions while $3 - a > 0$, ONE solution ($x = 4$) when $3 - a = 0$, none for $a > 3$. Tally: 4 solutions for $0 \leq a < 3$; three occur only when the second branch collapses to a point: $a = \mathbf{3}$. Check: $||x - 4| - 3| = 3$ gives $|x - 4| = 6$ (roots -2 and 10) or $|x - 4| = 0$ (root 4) — exactly three. An odd solution count for a symmetric equation always signals a degenerate branch at the center of symmetry.

PROBLEM 2 • MAXIMUM UNDER A LINEAR CONSTRAINT

Real numbers x and y satisfy $2x + 3y = 12$. What is the greatest value of the product xy ?

→ The dead end: try “nice” pairs like $(3, 2)$ or $(6, 0)$ and hope the maximum is among them — a search proves nothing.

The trick: substitute the constraint and reduce to one variable. From the relation, $x = (12 - 3y)/2$, so $xy = \frac{(12 - 3y)y}{2} = \frac{12y - 3y^2}{2}$ — a downward parabola, maximized at its vertex. Vertex: $y = 12/(2 \cdot 3) = 2$, whence $x = 3$ and $xy = \mathbf{6}$. No calculus — just the vertex of a parabola. A pleasing cross-check: at the maximum, $2x = 3y = 6$ — the linear “budget” splits evenly between the two terms, a general fact worth remembering as an instant answer check.